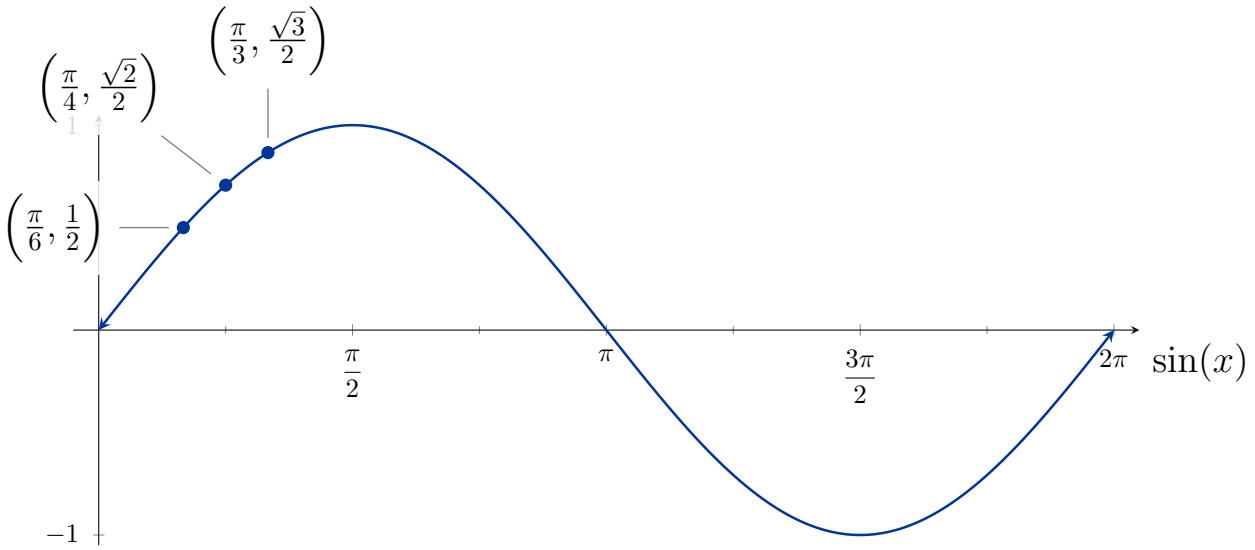
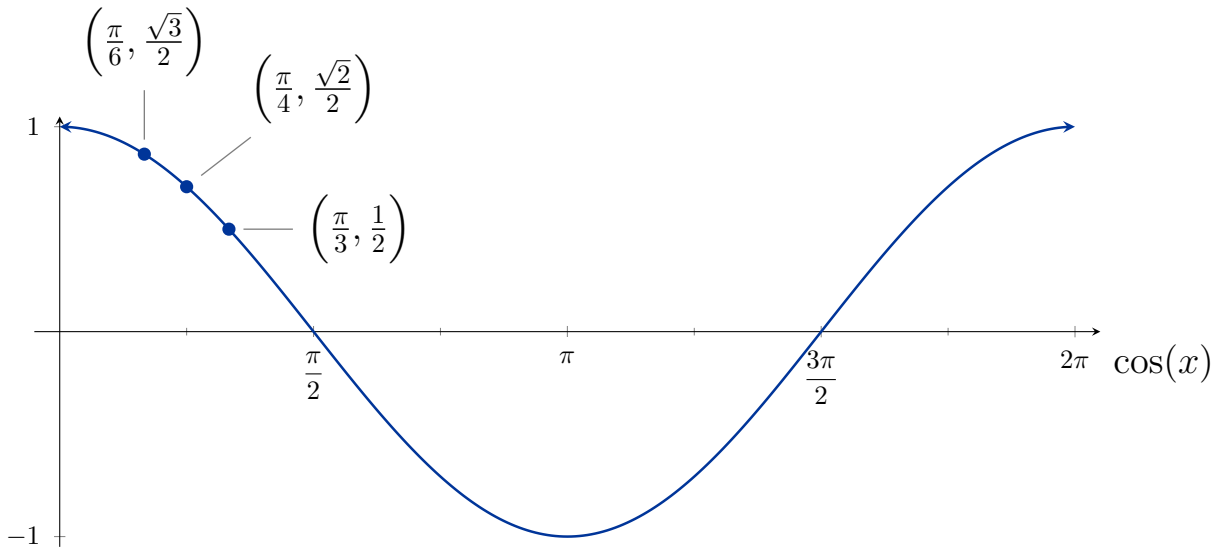
6.1: Graphs of the Sine and Cosine Functions

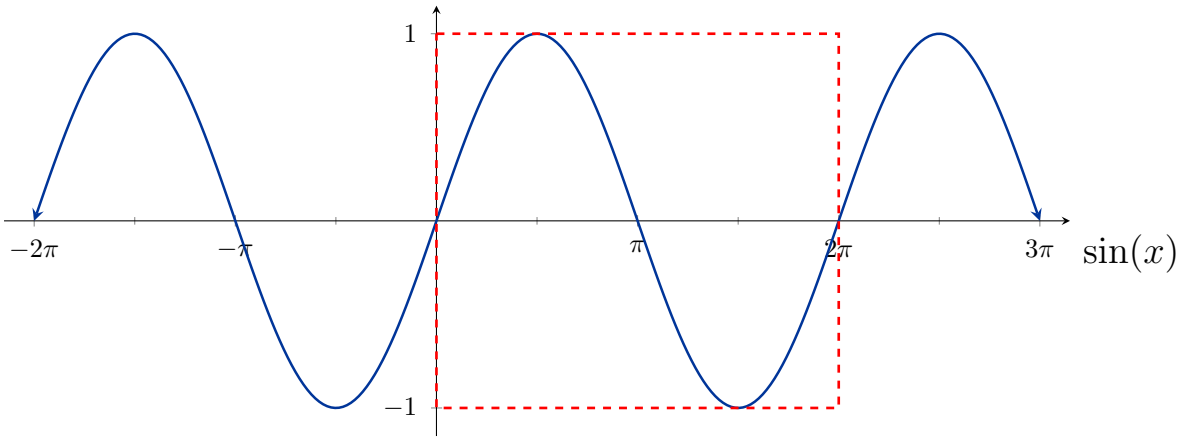


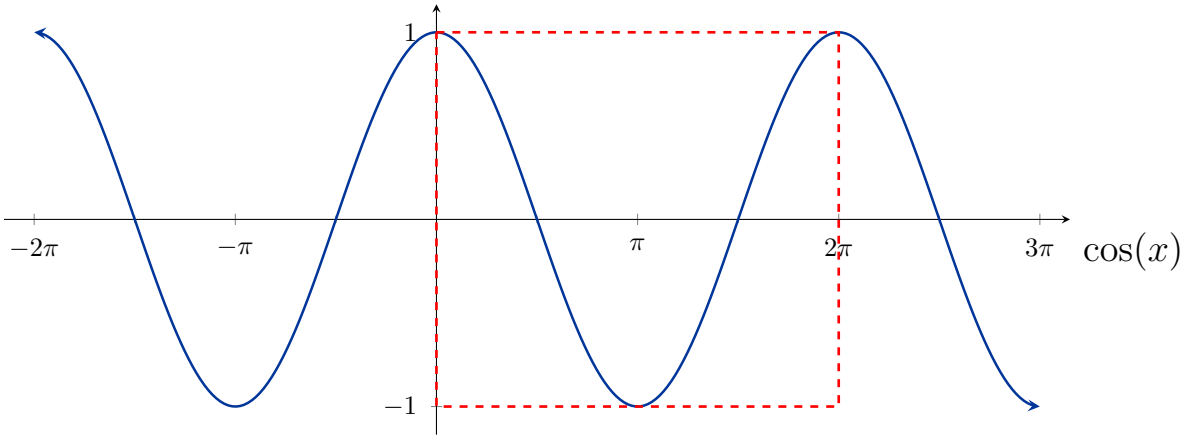


Animated graphs

Recall that

- $\sin(\theta)$ and $\cos(\theta)$ have period 2π,
- $\sin(\theta)$ is an odd function, and
- $\cos(\theta)$ is an even function:





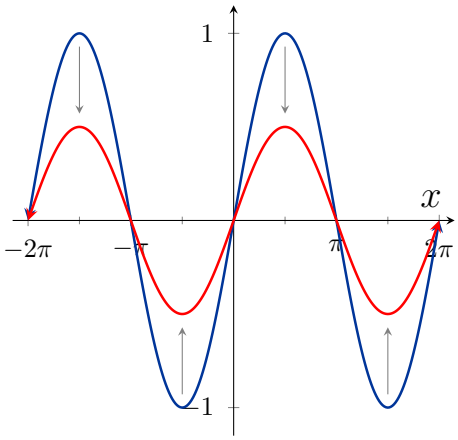
Recall from transformations of functions:

$$af(bx-h)+k$$

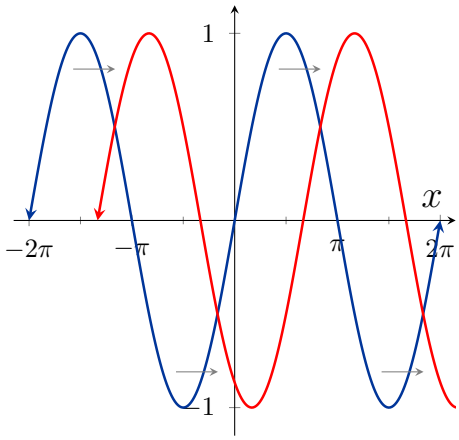
We have that $f(x)$ is

- stretched vertically by a
- stretched horizontally by $1/b$
- shifted right by h
- shifted vertically by k

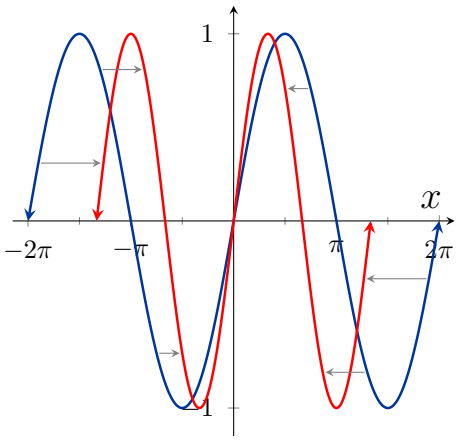
$\sin(x)$ vs. $\frac{1}{2}\sin(x)$



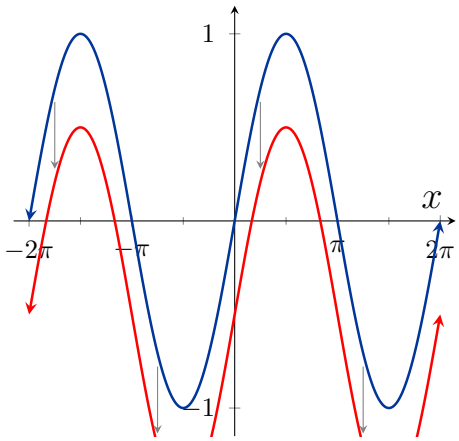
$\sin(x)$ vs. $\sin\left(x-\frac{2\pi}{3}\right)$



$\sin(x)$ vs. $\sin\left(\frac{3x}{2}\right)$



$\sin(x)$ vs. $\sin(x)-\frac{1}{2}$



Definition. (Sinusoidal functions)
A **sinusoidal function** is a function that has the same general shape as a sine or cosine function. The general forms of sinusoidal functions are

$$y = A\sin(Bx - C) + D \quad \text{and} \quad y = A\cos(Bx - C) + D,$$

where

- A determines the amplitude,
- B determines the period, $\longrightarrow \frac{2\pi}{B}$
- C determines the phase shift, and $\longrightarrow \frac{C}{B}$
- D determines the vertical shift.

Example. Identify the amplitude, period, phase shift, and vertical shift of

$$3\sin(2x) + 1$$
$$A \quad B \quad D$$

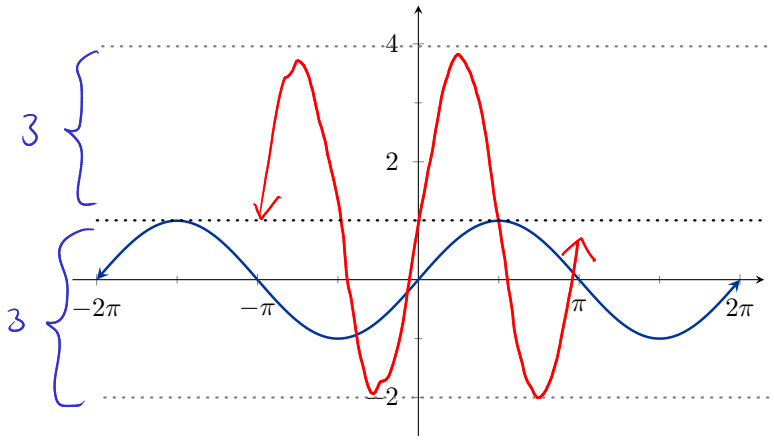
and sketch the graph:

Amplitude: 3

Period: $\frac{2\pi}{2} = \pi$

Phase shift: 0

Vertical shift: 1



Example. Identify the amplitude, period, phase shift, and vertical shift of

$$\frac{1}{2}\cos\left(\frac{x}{3}-\frac{\pi}{3}\right) \quad = \frac{1}{2}\cos\left(\frac{1}{3}(x-\pi)\right)$$

A B C

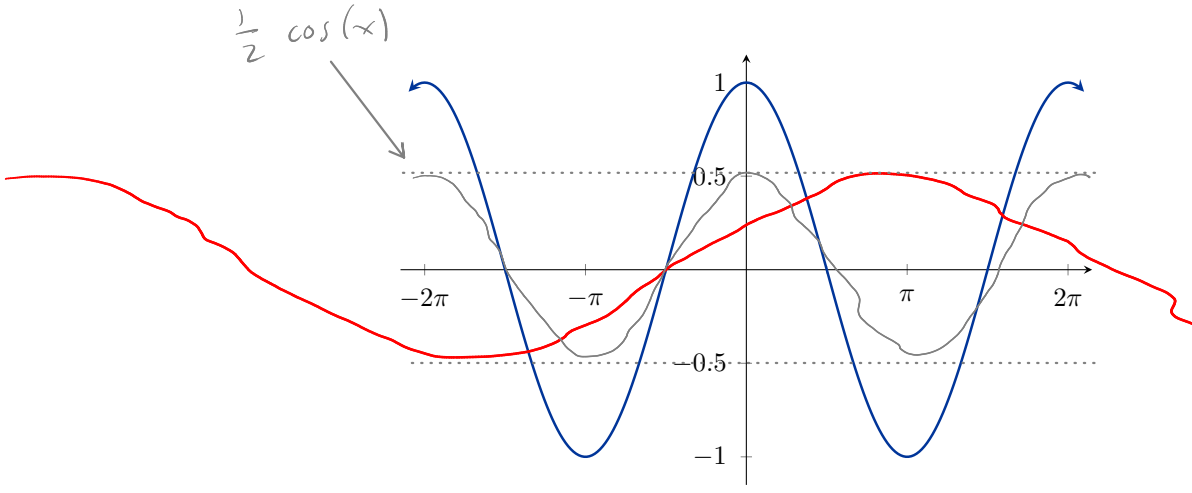
and sketch the graph:

Amplitude: $1/2$

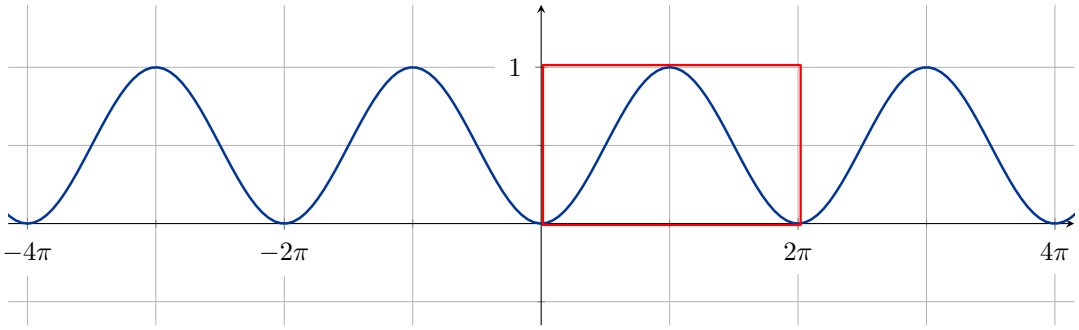
Period: $\frac{2\pi}{1/3} = 6\pi$

Phase shift: $\frac{1/3}{\pi/3} = \pi$

Vertical shift: 0



Example. Determine an equation for the following sinusoidal function:



Amplitude: $\frac{1-0}{2} = \frac{1}{2} \Rightarrow A = -\frac{1}{2}$

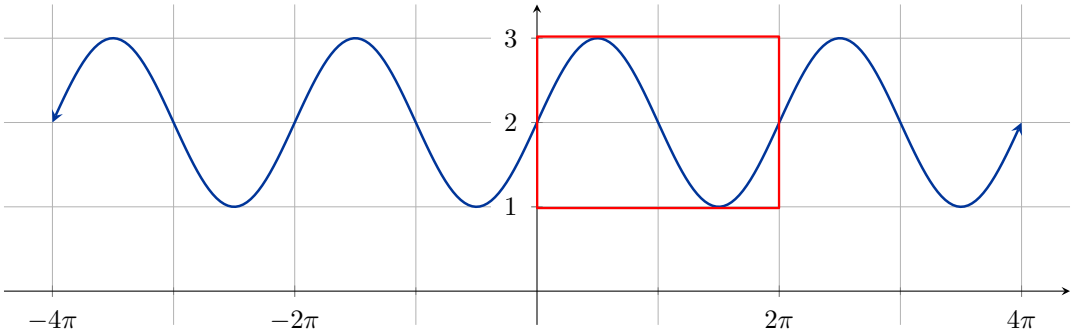
Period: $2\pi \Rightarrow B = 1$

Phase shift: $0 \Rightarrow C = 0$

Vertical shift: $1/2 \Rightarrow D = 1/2$

$\Rightarrow -\frac{1}{2}\cos(x) + \frac{1}{2}$

Example. Determine an equation for the following sinusoidal function:



Amplitude: $\frac{3-1}{2} = 1 \Rightarrow A = 1$

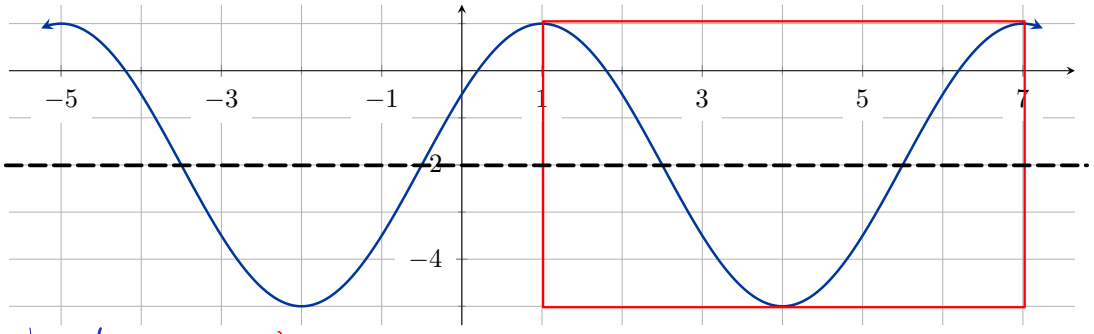
Period: $2\pi \Rightarrow B = 1$

Phase shift: $0 \Rightarrow C = 0$

Vertical shift: $2 \Rightarrow D = 2$

$\Rightarrow \sin(x) + 2$

Example. Determine an equation for the following sinusoidal function:



Amplitude: $\frac{1-(-5)}{2} = 3 \Rightarrow A = 3$

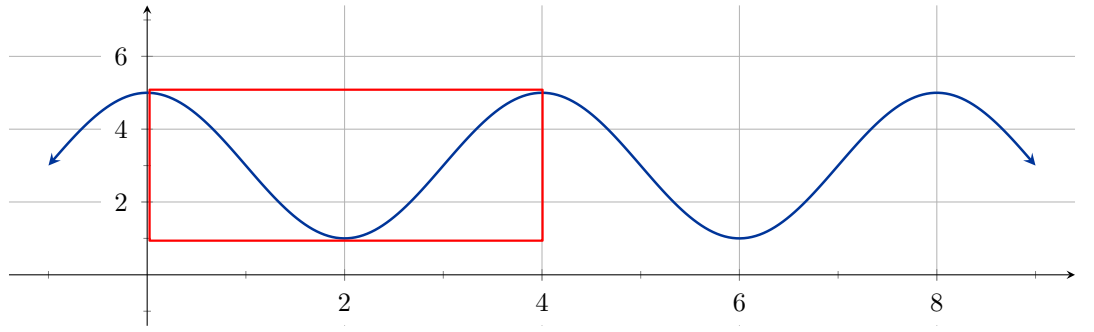
Period: $6 \Rightarrow B = \frac{2\pi}{6} = \frac{\pi}{3}$

Phase shift: $1 \Rightarrow C = \frac{\pi}{3}$

Vertical shift: $-2 \Rightarrow D = -2$

$$\Rightarrow 3\cos\left(\frac{\pi}{3}x - \frac{\pi}{3}\right) - 2$$

Example. Determine an equation for the following sinusoidal function:



Amplitude: $\frac{5-1}{2} = 2 \Rightarrow A = 2$

Period: $4 \Rightarrow B = \frac{2\pi}{4} = \frac{\pi}{2}$

Phase shift: $0 \Rightarrow C = 0$

Vertical shift: $3 \Rightarrow D = 3$

$$\Rightarrow 2\cos\left(\frac{\pi}{2}x\right) + 3$$

Alternatively

Phase shift: $1 \Rightarrow C = \frac{B}{1} = \frac{\pi}{2}$

$$\Rightarrow \boxed{2\sin\left(\frac{\pi}{2}x + \frac{\pi}{2}\right) + 3}$$